Azure Local: Day-2 Operations Runbook

The recurring operations for a running Azure Local instance: cluster updates, fault diagnosis, capacity management and the virtual machine lifecycle. Each procedure states its inputs, the steps, the evidence that proves it worked, and when to stop and escalate.

Status: prepared 2026-10-06 against Microsoft Learn (2609 release family). Commands are shown for orientation; run them with the account and context your environment requires, and test them on a non-production instance first. Some behaviour depends on your release and hardware; the text says where.

Principles

- **Read before you change.** Every procedure starts with a health check and ends with evidence.
- **One fault at a time.** Never stack a second change on a cluster that has an unresolved fault.
- **Recovery is proven, not assumed.** A cleared alert is not recovery. Recovery is the repair finished and redundancy restored.
- **Two update planes, two owners.** The cluster platform and the guest operating systems are updated by different tools.

1. Cluster updates

What is updated: the solution as one update train: the operating system, agents and services, and the OEM solution extension (drivers and firmware), orchestrated by Lifecycle Manager. **Cadence:** monthly quality updates, quarterly baseline updates, hotfixes as needed, and vendor extension updates. Stay within six months of the most recent release to remain supported. Feature releases can arrive about a week after Microsoft's release because the hardware vendor validates them.

Inputs: a maintenance window; a healthy cluster; a plan for what you do if the update fails. A failed update run can be resumed once the cause is fixed or accepted: select **Try again** on the failed stage in the portal, or run `Start-SolutionUpdate` on the failed update (add `-IgnoreWarnings` only when the blocking checks are in a Warning state and you accept them). After a power loss, restore power, run a system health check and then resume. If a node has a critical failure, repair the node first and then resume. If a retry does not resolve it, collect the diagnostic logs and open a support ticket.

Step	Portal	PowerShell on a node, as the deployment user
1. Confirm version and health	Update Manager, Azure Local	`Get-SolutionUpdateEnvironment` and check `CurrentVersion` and `HealthState`
2. Discover	Update Manager shows Updates available	`Get-SolutionUpdate | Where-Object State -like 'Ready*'`
3. Prepare (recommended)	Select Prepare to download and run readiness checks	`Get-SolutionUpdate -Id <id> | Start-SolutionUpdate -PrepareOnly`
4. Fix readiness failures	Check readiness, follow the remediation links	Read `HealthCheckResult`, follow `Remediation`
5. Install	Install now	`Get-SolutionUpdate -Id <id> | Start-SolutionUpdate`

Step	Portal	PowerShell on a node, as the deployment user
6. Verify	History shows success	`Get-SolutionUpdateEnvironment` shows the new version
7. Hardware	Included when the solution extension is part of the train	Apply any separate vendor update through the vendor's supported route

Expected duration: about 17 minutes of health checks and about 4 hours to install on one node, about 22 minutes and about 6 hours on four nodes. Content, load and hardware change this. Third-party tools for installing updates are not supported.

Stop and escalate when: health checks report Critical findings you cannot remediate from the links; an update stays in progress with no change for a long period; any node does not rejoin.

2. Fault diagnosis

Every fault uses the same five steps: **symptom, signal, differential diagnosis, action, recovery evidence.**

2.1 Drive failure

Symptom	A drive alert, a pool or volume in a Warning or Degraded state
Signal	Azure Monitor alert; physical disk `HealthStatus` or `OperationalStatus`
Diagnosis	Is the drive failed media, lost communication, or an administrative action? Is a repair job already running? Is another fault present?
Action	Storage Spaces Direct retires and evacuates a failed drive automatically; replace the drive physically; if the new drive is not added, check whether AutoPool is enabled
Recovery evidence	The storage repair job finished, all volumes are `Healthy`, redundancy is restored

```
Get-PhysicalDisk | Format-Table FriendlyName, SerialNumber, HealthStatus, OperationalStatus,
Usage
Get-StorageJob
Get-VirtualDisk | Format-Table FriendlyName, HealthStatus, OperationalStatus
# AutoPool setting (when a replacement drive is not pooled)
Get-StorageSubsystem Cluster* | Get-StorageHealthSetting | Select-Object
"System.Storage.PhysicalDisk.AutoPool.Enabled"
```

On a two-node, two-way-mirror cluster one fault leaves no redundancy until the repair completes. Reserve capacity (section 3) allows an immediate in-place repair before the drive is replaced.

2.2 Network intent drift

Symptom	An intent shows `ConfigurationStatus` other than `Success`, or a drift alert
Signal	`Get-NetIntentStatus`; Network ATC admin event log; Monitor alert
Diagnosis	Which setting differs from the intent, and who changed it? Network ATC owns the adapter properties it manages
Action	Let Network ATC remediate; if it does not converge, review the event log and follow the documented procedure for the change you need

Recovery evidence	Every intent reports `Success` on every node

```
Get-NetIntentStatus | Format-List Host, IntentName, ConfigurationStatus, ProvisioningStatus
```

Configuration drift is a fault. The fix is to change the intent, not the adapter.

2.3 Node failure

Symptom	A node is down; VMs restart on the surviving node; an alert fires
Signal	Cluster node state; Monitor alert; VM state
Diagnosis	Is it the node, its network, or its power? Is the witness reachable? On two nodes, one node plus the witness keeps quorum; losing both a node and the witness takes the cluster offline
Action	Restore the node through the hardware management path; do not stack another change; when the node returns, let storage resync
Recovery evidence	The node is `Up`, storage repair finished, volumes healthy, VMs on their intended owners

3. Capacity management

What "full" means: on a two-way mirror, usable capacity is about half of raw, and a node loss must still leave room for the VMs and the repair.

1. Compute the healthy-state usage and the usage with one node unavailable. The second number is the one that limits you.
2. Keep reserve capacity: leave unallocated pool capacity equal to one capacity drive per server, up to four (this is free space, not spare drives), so a failed drive can repair in place.
3. Decide with numbers. If an additional workload breaks the node-loss reserve, reject it or expand first; record the trigger for expansion.
4. Expansion: adding a node changes resiliency options and the network intents; plan it as a change, not an event. Rack Aware clusters expand in pairs.

```
Get-StoragePool -IsPrimordial $false | Format-List FriendlyName, Size, AllocatedSize
Get-VirtualDisk | Format-Table FriendlyName, Size, FootprintOnPool
```

4. Virtual machine lifecycle through Azure Arc

Day 2 for workloads is Azure, not a console session. Create, govern and delete virtual machines from Azure, so that RBAC, policy, monitoring and update management apply.

Prerequisites: an image (from the marketplace, a storage account, or a local share), a logical network, a custom location, and a storage path on a workload volume (never the infrastructure volume).

```
az stack-hci-vm create --name <vm> --resource-group <rg> --custom-location <custom location id>
\
  --image <image name> --admin-username <admin> --admin-password <from a secure prompt> \
  --nics <nic name> --storage-path-id <storage path id>
```

Step	Check
Create	The VM lists as Succeeded under Azure Local virtual machines
Govern	Tags set; policy compliance evaluated; the VM appears in Update Manager and monitoring scopes
Protect	A backup policy is assigned; replication enabled if the tier requires it
Delete	Remove associated resources, then the VM; deleting an Azure Local VM after a failover to Azure needs manual clean-up

A VM disconnected from Azure Arc can be reconnected within the Arc agent's 45-day window.

5. Routine schedule

Cadence	Task
Daily	Review Monitor alerts and the Insights health view
Weekly	Check backup jobs and replication health; review policy compliance
Monthly	Review and apply the quality update in the maintenance window
Quarterly	Apply the baseline update; run a DR test failover; review capacity and expansion triggers; check certificate and secret expiry (the local administrator account and secrets are yours to rotate)
After any change	Re-run the readiness checks